MINISTRY OF EDUCATION AND TRAINING
CAN THO UNIVERSITY

SUMMARY REPORT RESEARCH PROJECT

- Project code: THS2026-68
- Project title: Development of an Edge AI-based System for Multimodal Analysis and Clustering of Flood Rescue Events
- Project period: May–October 2026
- Total cost: 15,000,000 VND
- Project leader: Lê Thị Ngọc Ảnh (0376932994)
- Project members: Nguyễn Thanh Trọng, Cao Tường Hưng, Nguyễn Như Quỳnh, Ngô Hưng Thịnh.
- Advisor: TS. Nguyễn Thanh Khoa (ntkhoa@ctu.edu.vn)
- Necessity of the project: Duplicate, unlocated or delayed rescue reports hinder coordination under intermittent connectivity.
- Objectives: Develop a prototype combining edge image inference, compact transmission and map-based event clustering.
- Methodology: Train MobileNetV3-Large and evaluate image classification, clustering, ranking and dispatch separately using saved artifacts and simulation datasets.
- Project activities: Image AI, structured urgency, persistent queues, adaptive sending, synchronization and dashboard. A translated-Vietnamese/AI-silver pilot runs NLP and offline fusion; native data, information extraction and camera/Android evaluation remain open.
- Research results: The original seed-1024 model achieves 75.78% accuracy and 74.26% macro-F1 on 256 CPU test images, with three critical errors. Quantization reduces size and accuracy. Phone benchmarks cover 32 configurations on four devices. The app, backend and dashboard were developed; five-flag LR is not free-text NLP and its score is not used by the ranking adapter. An English CrisisMMD fusion pilot reaches 78,35% macro-F1; effect CIs include zero. A translated-Vietnamese/AI-silver pilot reaches 64,91% frozen-fusion macro-F1 and 33,33% request recall; independent human gold and application fusion remain open.
- Research new finding: RQ1 does not establish Product superiority; RQ2 does not establish general alignment gains and retains three raw Spearman and five Holm replay differences. RQ3 favors nearest-first over revised priority on simulated harm and deadline misses. Field rescue benefits remain unmeasured.
- Application potentials: Research and workflow demonstration. Urgency integration, mobile provenance, repeated device/network/SMS tests and expert validation remain necessary.